

1)insert at beginning:

```
#include <stdio.h>
```

```
#include<stdlib.h>
```

```
struct node{  
    int data;  
    struct node *next;  
}*head=NULL;
```

```
void append(int num){  
    struct node *newnode=(struct node*)malloc(sizeof(struct node));  
    newnode->data=num;  
    newnode->next=NULL;  
    if(head==NULL){  
        head=newnode;  
    }  
    else{  
        struct node *temp=head;  
        while(temp->next!=NULL){  
            temp=temp->next;  
        }  
        temp->next=newnode;  
    }  
}
```

```
void insatbeg(int a){  
    struct node *newnode=(struct node*)malloc(sizeof(struct node));  
    newnode->data=a;  
    newnode->next=head;  
    head=newnode;  
}
```

```

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);
}

int main()
{
    int n,num,a;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }
    scanf("%d",&a);
    insatbeg(a);
    display();

    return 0;
}

```

2)insert at position:

```
#include <stdio.h>
```

```
#include<stdlib.h>
```

```

struct node{
    int data;
    struct node *next;
}*head=NULL;

```

```

void append(int num){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=num;
    newnode->next=NULL;
    if(head==NULL){
        head=newnode;
    }
    else{
        struct node *temp=head;
        while(temp->next!=NULL){
            temp=temp->next;
        }
        temp->next=newnode;
    }
}

void insatbeg(int a){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=a;
    newnode->next=head;
    head=newnode;
}

void insatpos(int pos,int a){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    struct node *temp=head;
    for(int i=0;i<pos-2;i++){
        temp=temp->next;
    }
    newnode->data=a;
    newnode->next=temp->next;
    temp->next=newnode;
}

```

```

}

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);
}

int main()
{
    int n,num,a,pos;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }
    printf("enter position:");
    scanf("%d",&pos);
    printf("enter value:");
    scanf("%d",&a);
    insatpos(pos,a);
    display();

    return 0;
}

```

3)delete at beginning:

```

#include <stdio.h>

#include<stdlib.h>

```

```

struct node{
    int data;
    struct node *next;
}*head=NULL;

void append(int num){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=num;
    newnode->next=NULL;
    if(head==NULL){
        head=newnode;
    }
    else{
        struct node *temp=head;
        while(temp->next!=NULL){
            temp=temp->next;
        }
        temp->next=newnode;
    }
}

void delatbeg(){
    struct node *temp=head;
    head=head->next;
    free(temp);
}

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
}

```

```

        printf("%d",temp->data);
    }
int main()
{
    int n,num,a,pos;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }

    delatbeg();
    display();

    return 0;
}

```

4)Delete at end:

```

#include <stdio.h>
#include<stdlib.h>

struct node{
    int data;
    struct node *next;
}*head=NULL;

void append(int num){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=num;
    newnode->next=NULL;
    if(head==NULL){

```

```

        head=newnode;
    }
    else{
        struct node *temp=head;
        while(temp->next!=NULL){
            temp=temp->next;
        }
        temp->next=newnode;
    }
}

void delatend(int n){
    struct node *temp=head;
    for(int i=0;i<n-2;i++){
        temp=temp->next;
    }
    struct node *last=temp->next;
    temp->next=NULL;
    free(last);
}

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);
}

int main()
{
    int n,num,a,pos;
    scanf("%d",&n);

```

```

for(int i=0;i<n;i++){
    scanf("%d",&num);
    append(num);
}

delatend(n);
display();

return 0;
}

```

5)delete at position:

```

#include <stdio.h>
#include<stdlib.h>

struct node{
    int data;
    struct node *next;
}*head=NULL;

void append(int num){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=num;
    newnode->next=NULL;
    if(head==NULL){
        head=newnode;
    }
    else{
        struct node *temp=head;
        while(temp->next!=NULL){
            temp=temp->next;

```



```

    }
    temp->next=newnode;
}
}

void delatpos(int pos){
    struct node *temp=head;
    for(int i=0;i<pos-2;i++){
        temp=temp->next;
    }
    struct node *del=temp->next;
    temp->next=temp->next->next;
    free(del);
}

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);;
}

int main()
{
    int n,num,pos;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }
    scanf("%d",&pos);
    delatpos(pos);
}

```

```
display();

return 0;
}
```

6)Search an element:

```
#include <stdio.h>
#include<stdlib.h>
```

```
struct node{
    int data;
    struct node *next;
}*head=NULL;
```

```
void append(int num){
    struct node *newnode=(struct node*)malloc(sizeof(struct node));
    newnode->data=num;
    newnode->next=NULL;
    if(head==NULL){
        head=newnode;
    }
    else{
        struct node *temp=head;
        while(temp->next!=NULL){
            temp=temp->next;
        }
        temp->next=newnode;
    }
}
```

```
void search(int a,int n){
    for(int i=0;i<n;i++){
```

```

        if(head->data == a){
            printf("element found on %d position",i+1);
            break;
        }
        head=head->next;
    }
}

```

```

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);
}

```

```

int main()
{
    int n,num,a;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }
    scanf("%d",&a);
    search(a,n);

    return 0;
}

```

7)Middle element:

```

#include <stdio.h>

```

```
#include<stdlib.h>
```

```
struct node{  
    int data;  
    struct node *next;  
}*head=NULL;
```

```
void append(int num){  
    struct node *newnode=(struct node*)malloc(sizeof(struct node));  
    newnode->data=num;  
    newnode->next=NULL;  
    if(head==NULL){  
        head=newnode;  
    }  
    else{  
        struct node *temp=head;  
        while(temp->next!=NULL){  
            temp=temp->next;  
        }  
        temp->next=newnode;  
    }  
}
```

```
void middle(int n){  
    for(int i=0;i<=n/2;i++){  
  
        if(i==n/2){  
            printf("middle element is %d\n",head->data);  
            break;  
        }  
        head=head->next;  
    }  
}
```

```

}

void display(){
    struct node *temp=head;
    while(temp->next!=NULL){
        printf("%d\n",temp->data);
        temp=temp->next;
    }
    printf("%d",temp->data);
}

int main()
{
    int n,num,a;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&num);
        append(num);
    }
    middle(n);

    return 0;
}

```